

LABLAB.AI · ASSEMBLYAI VOICE AGENT HACKATHON

# VoiceRemitNG

Nigeria remittance voice agent

Speak → AssemblyAI Realtime STT → remittance tools → clear guidance

Quantumwoof · Nigeria · MIT License

# The problem

- Diaspora sends USD → family in Nigeria needs ■ for rent, school, hospital.
- Opaque fees vs FX markups across Wise, Remitly, Sendwave, Western Union.
- Confusion between CBN / parallel / app retail USD→NGN rates.
- Bad timing (weekends / after Lagos banking hours) delays bank payouts.
- Failed transfers from wrong NUBAN, name mismatch, or incomplete cash-pickup IDs.
- Most “help” is blogs or text chatbots — people already talk by voice.

## *Voice-shaped problem:*

meet people where they speak · Nigerian accents / Pidgin code-switch · practical tools after STT

# Solution — VoiceRemitNG

1

## Hear

Mock STT in DEMO  
AssemblyAI Realtime  
STT in LIVE mode

2

## Route

Intent → tools  
fee · FX · timing  
receive checklist

3

## Answer

Spoken-style advice  
+ structured details  
tables / checklists

4

## Judge-ready

Streamlit + CLI  
DEMO\_MODE=1  
No API key needed

Mocks are hackathon demos, not live quotes. VoiceRemitNG advises — it does not send money.

# AssemblyAI usage

## DEMO

DEMO\_MODE=1 — Mock STT (typed / canned). No key. Judges run offline.

## LIVE Realtime

assemblyai.streaming.v3.RealTimeTranscriber → wss://streaming.assemblyai.com/v3/ws

## LIVE file

Prerecorded / Universal transcription for Streamlit audio upload

## Speech model

universal-streaming-multilingual (accents + English↔Pidgin). Override via env.

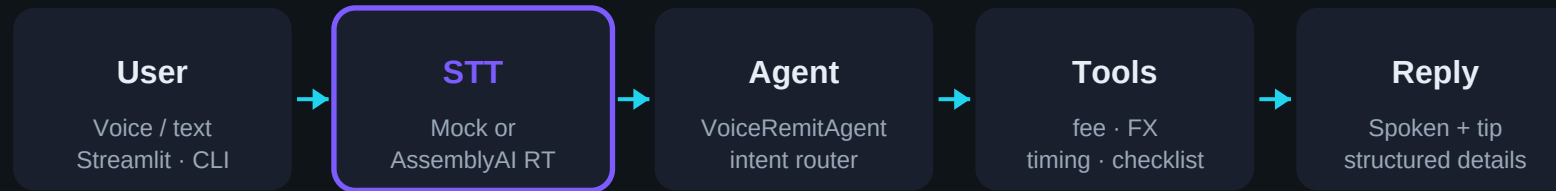
## Honest scope

Realtime STT is the wired voice surface; remittance tools stay local Python.

## Next step

Same tools can register as AssemblyAI Voice Agent tools without rewriting product logic.

# Architecture



DEMO: MockSTT · LIVE: Realtime STT + optional mic / file upload

## Intents

fee\_compare

fx\_check

send\_now\_advice

receive\_checklist

help

# DEMO MODE tools & setup

## Remittance tools

### fee\_compare

All-in cost across corridors; ranks by ■ received

### fx\_check

Mock USD/NGN mid + Lagos time + retail-rate tip

### send\_now\_advice

Banking hours / weekend guidance (Africa/Lagos)

### receive\_checklist

Bank / wallet / cash-pickup KYC & NUBAN

## Quick start

- 1 `git clone .../voice-remit-ng`
- 2 `python3 -m venv .venv && source .venv/bin/activate`
- 3 `pip install -r requirements.txt`
- 4 `DEMO_MODE=1 python smoke_test.py`
- 5 `DEMO_MODE=1 streamlit run app.py`
- 6 `DEMO_MODE=1 python cli.py "Compare fees..."`

Live: `DEMO_MODE=0 + ASSEMBLYAI_API_KEY` → Realtime STT / file upload / optional `--mic`  
Smoke test exits 0 offline — judges can verify without a key.

# Team & submit notes

**Speak remittances. Confirm in-app. Don't ship money.**

AssemblyAI Realtime STT · Nigeria remittance tools · DEMO offline

[github.com/Quantumwoof/voice-remit-ng](https://github.com/Quantumwoof/voice-remit-ng)

Team: Quantumwoof · MIT License · lablab AssemblyAI hackathon

Assets: VoiceRemitNG-cover.png · VoiceRemitNG-slides.pdf · VoiceRemitNG-demo.mp4

Rates/fees are mocks — not financial advice. VoiceRemitNG does not move money.